ZONGQI CUI

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RESEARCH INTERESTS

Bayesian cognitive modeling and large language models as approximations of human inference; theory of mind and intention attribution from dialogue; distillation of structured reasoning into smaller, auditable models; calibration and interpretability of LLM-based agents.

EDUCATION

Emory University, M.S. in Computer Science Atlanta, GA · 08/2024 – 05/2026

- GPA: 3.64 / 4.0
- Relevant coursework: Machine Learning (A), Natural Language Processing (A), Artificial Intelligence (A), Computer Vision (A), Spatial Computing (A), Deep Learning on Graphs (A−), Algorithms (A−), Python Programming (A−), Data Mining (B+).

Univ. of Shanghai for Science & Technology, B.S. Computer Science Shanghai · 08/2020 – 06/2024

- GPA: 3.52 / 4.0 (86.69 / 100)

PUBLICATIONS & PREPRINTS

- Cui, Z., Lin, B. *Distilling Bayesian Belief States into Language Models for Auditable Negotiation.* Submitted to NeurIPS 2026 (under review). [arXiv link] · code

RESEARCH EXPERIENCE

Research Assistant — Bayesian Belief Distillation for Negotiation Agents 11/2025 – 05/2026

First-author research · Bytes of Minds Lab, Icahn School of Medicine at Mount Sinai · Advisor: Prof. Baihan Lin · NeurIPS 2026 submission (under review)

- Designed a "language as likelihood, Bayes as update" framework: a Llama-3.3-70B teacher scores dialogue compatibility against six opponent priority orderings to produce calibrated, turn-level posteriors over hidden preferences in CaSiNo campsite negotiation.
- Distilled the Bayesian teacher into a Llama-3.1-8B student via LoRA SFT; the student emits auditable tagged outputs (posterior, intent, content, utterance) and preserves belief calibration on 1,054 held-out turns (Brier 0.114 vs. teacher 0.085, vs. 0.139 uniform reference).
- Established a same-scale supervised baseline that exceeds prior CaSiNo SOTA (Chawla et al. 2022): k-penalty Top-1 76.2 ± 1.0 under matched 5-fold CV, vs. 70.0 for RoBERTa+CD+CA+DND.
- Diagnosed and reported pre-registered null results — SVO swap-test insensitivity within fixed λ regimes, Big Five style-gate failure (χ^2 $p = 0.998$) — separating λ-scale effects from social-label assignment effects.
- Built belief-policy decomposition, posterior-trajectory visualizations, and posterior-prefix interventions to make weak belief-action coupling visible rather than hidden.
- Stack: Llama-3.1 / 3.3, LoRA & QLoRA via TRL, PyTorch, HPC (LSF on Arion / Minerva, A100 / H100).

Drug Efficacy Modeling and LLM-Based Explainability 02/2026 – 05/2026

Industry-supervised research · SK Life Science

- Reproduced and extended a Data-driven Explanatory Model (DEM) using Random Forest on 311K drug–cell-line pairs across 2,024 multi-omic features (RNA-seq, mutations, molecular fingerprints from CTRP / CCLE / PubChem); achieved $R^2 \approx 0.66$ and Pearson $r \approx 0.82$ on held-out CTRP data.
- Engineered a custom perturbation kernel based on Random Forest leaf-agreement with weighted tree logic and depth-aware scaling to model drug-tumor similarity for downstream treatment-response inference.

- Built an end-to-end SHAP-to-LLM explainer: extracted TreeSHAP attributions, mapped features to biomedical metadata (SMARTS substructures, gene-expression z-scores, drug targets via PubChem / DGIdb), and assembled structured prompts for grounded natural-language explanations of IC50 predictions.
- Benchmarked four candidate LLMs (Llama 3.1 8B, Qwen 2.5 32B, BioMistral 7B, Meditron 7B) on explanation quality; general-purpose models outperformed biomedically fine-tuned ones, consistent with Dorfner et al. (2024). Designed a QLoRA fine-tuning pipeline targeting H100 GPUs with GGUF export for local inference.
- Co-designed a blinded expert-evaluation framework: 25 stratified drug-cell pairs × 4 explanation variants (full-feature, SHAP-only, shuffled-SHAP, wrong-drug) scored via forced-choice paired comparison with win-rate aggregation and inter-rater agreement.

Research Assistant — Modeling Human Strategic Behavior with Decision Transformers 09/2025 – 12/2025

Bytes of Minds Lab, Icahn School of Medicine at Mount Sinai · Advisor: Prof. Baihan Lin

- Applied a Decision Transformer (GPT-style sequence model with reward-conditioned trajectories) to predict human cooperation behavior in the Iterated Prisoner's Dilemma, reframing sequential strategic choice as offline RL.
- Designed state–action–return representations and return-to-go conditioning, evaluated against Transformer and BERT baselines on action-prediction accuracy, cooperation-rate trajectories, and behavioral consistency.
- Applied SHAP to interpret learned policies; identified return-to-go and recent-opponent-action features as dominant drivers of cooperation. Code: github.com/kaneis1/decision.

Graph-Augmented Multimodal Representation Learning with CLIP 03/2025 – 05/2025

Independent research

- Developed a relation-guided contrastive learning model combining CLIP feature extraction with graph neural networks to align image and text embeddings within a multimodal product graph built from Amazon co-purchase data.
- Designed cross-modal and inter-node contrastive objectives leveraging edge semantics for zero-shot node classification and link prediction.
- Outperformed CLIP, DeCLIP, UniCL, SigLIP, and ImageBind baselines, with a +30% average improvement in Hit@5. Stack: PyTorch, HuggingFace Transformers, NetworkX, CUDA.

Detecting Mild Cognitive Impairment from Verbal Fluency 02/2025 – 04/2025

Collaborative research — language as a window into latent cognitive state

- Analyzed 650+ transcribed audio samples from the B-SHARP dataset across T-word, S-word, Animal, and Fruit fluency tasks; extracted both classical features (word diversity, switching, clustering) and LLM-derived semantic features.
- Applied Mann–Whitney U and t-tests with FDR correction, Random Forest, and logistic regression to evaluate statistical significance and predictive performance.
- Integrated Whisper v3, HuggingFace Transformers, GPT-4o, and scikit-learn into a unified pipeline for transcription, feature mining, and classification; led feature extraction, GPT-4o annotation, and error analysis.

MBTI Personality Prediction with BERT 04/2025

Independent research — early exploration of inferring latent traits from text

- Designed and fine-tuned a sentence-level MBTI classification system on Kaggle user-generated text using HuggingFace Transformers and PyTorch — a precursor to my later Bayesian belief-distillation work on inferring hidden mental states from dialogue. Code: github.com/kaneis1/Psy.

INDUSTRY EXPERIENCE

Computer Vision Intern, Bayer Cropscience China Co., Ltd. Shanghai · 08/2023 – 11/2023

- Adapted YOLOv5 for tomato detection and tracking in field imagery; tuned anchors and augmentation to improve recall on small-object cases.
- Implemented and tested algorithm variants in Python, integrated them into the team's data and ML pipelines.

TECHNICAL SKILLS

- **Programming:** Python (PyTorch, HuggingFace Transformers, TRL, PEFT, scikit-learn, pandas), C++, SQL, LaTeX.
- **ML & Research:** LoRA / QLoRA fine-tuning, knowledge distillation, Bayesian inference, calibration metrics (Brier, NDCG, k-penalty), TreeSHAP, contrastive learning, prompting and structured-CoT evaluation.
- **Infrastructure:** HPC clusters (LSF / Slurm; Arion, Minerva), multi-GPU training (A100, H100), GGUF export for local inference, Git, Overleaf.
- **Languages:** Mandarin (native), English (professional working).

HONORS & ACTIVITIES

- **Competitive Programming (ICPC / CCPC), 2018 – 2024.** Six years on the University ACM team; 1,000+ problems solved across graphs, dynamic programming, string matching, and computational geometry, with consistent regional-contest participation.